

MARCUS BENETT ZAENS

sneazmbp@gmail.com | marcus-zaens.github.io

EDUCATION

Plano East Senior High School

International Baccalaureate Diploma | High School Diploma

Aug. 2023 - Jun. 2027

Plano, TX

St. Mark Catholic School

Honors Program | Grades 5-8

Aug. 2019 - May 2023

Plano, TX

St. Catherine of Siena School

Grades 1-4

Sep. 2015 - Jun. 2019

Manchester, NH

RELEVANT COURSEWORK

Computer Science IB HL Computer Science, AP Computer Science Principles

Mathematics IB HL Math Analysis & Approaches, AP Precalculus, Algebra 1 & 2 Honors, Geometry Honors

Physics & Engineering IB SL Physics, PLTW Engineering Science

PROJECTS

Custom Mechanical Keyboard | github.com/mbzgr8/keyboard-firmware

- Designed and routed a custom PCB in KiCAD, implementing a diode-protected key matrix
- Soldered diodes, mechanical switches, and an RP2040 Pro Micro microcontroller
- Programmed the RP2040 Pro Micro to support QMK firmware and VIA compatibility with custom keymaps
- Created and 3D printed a custom 4-part keyboard case in Fusion 360

Detecting Myopia with Logistic Regression and CNN Models in Python

- Engineered logistic regression and convolutional neural network approaches for classifying myopia from fundus images
- Built a logistic regression baseline achieving 87.6% accuracy, validating with a confusion matrix
- Designed a multi-layer CNN architecture combining convolutional, max-pooling, and dense flattening layers
- Trained a CNN on 1,000 labeled fundus images using TensorFlow and Keras, reaching 87.35% accuracy
- Preprocessed and augmented data via flipping, rotating, and zooming with a 70/30 train-test split

Endless Runner Game with Computer Vision

- Created an arcade-style endless runner game controlled via live facial tracking using MediaPipe and OpenCV
- Developed the game in Python with Pygame, including score tracking, high-score saving, and a home screen
- Implemented procedural obstacle generation, dynamic difficulty scaling, and live camera control
- Designed custom pixel-art sprites, graphics, and animations with Pixilart

Predicting House Prices with Regression & Neural Networks

- Built linear, multiple, and polynomial regression models and a neural network on the Ames Housing dataset
- Created in Python using pandas, NumPy, TensorFlow, and scikit-learn libraries
- Implemented a 70/30 train-test split and polynomial features to improve model fit
- Performed exploratory data analysis using Matplotlib, including scatter plots, histograms, and bar graphs
- Evaluated model performance using mean squared error (MSE)

4-DOF Robotic Arm

- Engineered a four degree-of-freedom robotic arm using an Arduino Uno Rev3, servo motors, and potentiometers
- Designed and 3D printed custom mechanical parts in Fusion 360
- Programmed motor control and motion sequences in C, achieving smooth, reliable, and repeatable movement

Personal Portfolio Website | marcus-zaens.github.io

- Built and deployed portfolio site using HTML, CSS, and JavaScript to showcase projects and technical skills
- Managed version control with Git and hosted the site on GitHub Pages

WORK EXPERIENCE

Math and Reading Student Instructor

Jun. 2023 - Present

Kumon North America, Inc.

Murphy, TX

- Tutored groups of 3-4 students in math and reading; improving comprehension, problem-solving skills, and confidence
- Graded homework, administered classwork, and managed student records
- Monitored student progress and adjusted lesson plans to ensure continuous improvement

EXTRACURRICULARS

Computer Science Club

Officer (2026 - Present), Member (2023 - 2026)

Math Club

Officer (2026 - Present), Member (2023 - 2026)

Chess Club

Co-Founder 2026 - Present

Environmental Safety Club

Distinguished Member (2026 - Present), Member (2025 - Present)

US Chess Federation

Member 2023 - Present

Pianist

2015 - Present

HONORS AND AWARDS

National Honors Society at Plano East Senior High School

Member (2023 - Present)

Mu Alpha Theta – National Mathematics Honor Society

Member (2023 - Present)

Best of Texas 11th Grade Computer Science, 12th Place (Round 1), 16th Place (Round 2)

May 2026

University Interscholastic League (UIL) Number Sense, 33rd Place

May 2026

University Interscholastic League (UIL) Mathematics, 32nd Place

May 2026

AP Scholar with Distinction

May 2026

University Interscholastic League (UIL) Number Sense, 30th Place

May 2025

AP Scholar with Honor

May 2025

Private Schools Interscholastic Association (PSIA), Math, 6th Place

May 2023

Catholic Math League, Math 7, 1st Place School

May 2022

Private Schools Interscholastic Association (PSIA), Math, 5th Place

May 2022

Selected to represent NH, FIRST LEGO League “Into Orbit” International Open Championship

May 2019

FIRST LEGO League “Into Orbit” State Championship, 2nd Place

Dec. 2018

LANGUAGES, SKILLS, AND CERTIFICATES

Languages English, Spanish, Tagalog

Programming Python, Java, C, JavaScript, HTML/CSS

Tools Git, GitHub, VS Code, Linux, Arduino, Breadboarding, Soldering

CAD & Design Fusion 360, Blender, KiCAD

Certificates IBM Artificial Intelligence Fundamentals, NOCTI Engineering Technology Foundations

COMMUNITY SERVICE AND VOLUNTEER WORK

International Baccalaureate Mentorship Program

Jan. 2023 - Present

- In partnership with the school's tutoring club, tutored IH students one-on-one in Algebra I, Geometry, and Algebra II
- Scheduled weekly meetings, provided live practice problems, and prepared students before major examinations

Love for Our Elders

2023 - Present

- Created unique pop-up letters for the elderly in isolation, spreading words of encouragement and connection

Environmental Safety (S.A.V.E.) Club

2025 - Present

- Participated in collaborative school sustainability efforts and renewable practices throughout the school year
- Engaged in eco-friendly projects, environmental advocacy, and hands-on initiatives to promote a sustainable future